

3.2.2 Micronutrients

3.2.2.1 Vitamins

Content	Students must know and understand	Suggested application and food preparation skills
Fat soluble • vitamin A • vitamin D • vitamin E • vitamin K	• the functions • main sources • effects of deficiency and excess • related dietary reference values.	Knife skills – fillet and slice fish and/or fruits and vegetables (S2).
Water soluble • B group – B1 (thiamin), B2 (riboflavin), B3 (niacin), folic acid, B12 • vitamin C (ascorbic acid) • loss of water soluble vitamins when cooking (B group and Vitamin C).	• the functions • main sources • effects of deficiency and excess • related dietary reference values • how preparation and cooking affects the nutritional properties of food.	• Cooking methods – water based using the hob – steaming, boiling, simmering and poaching (S6). • Knife skills – cut fruit and vegetables into even size pieces (ie batons, julienne) (S2).
Antioxidant functions of vitamins • vitamin A • vitamin C • vitamin E.	The role of antioxidants in protecting body cells from damage.	Preparing fruit and vegetables eg making different salads inclusive of vegetables, nuts or eggs which contain antioxidant vitamins (S2/S3).

3.2.2.2 Minerals

Content	Students must know and understand	Suggested application and food preparation skills
- calcium - iron - sodium (salt) - fluoride - iodine - phosphorus.	- the functions - main sources - effects of deficiency and excess - related dietary reference values.	- Preparing vegetables, meats or alternatives which are high in iron (S2). - Preparing dairy foods, which are high in calcium, for example when making a white sauce (S8). - Reducing the salt in recipes eg when tasting and seasoning, replace salt with herbs and spices.

3.2.2.3 Water

Content	Students must know and understand	Suggested application and food preparation skills
The importance of hydration and the functions of water in the diet.	- functions of water to eliminate waste from the body, cooling and for digestion. - how water is lost from the body. - how much water/fluid is needed each day. - occasions when extra fluids are needed.	Preparing fruit and vegetables, for example in soup making – scissor snip, crush, grate, peel. Make juices and smoothies (S3).